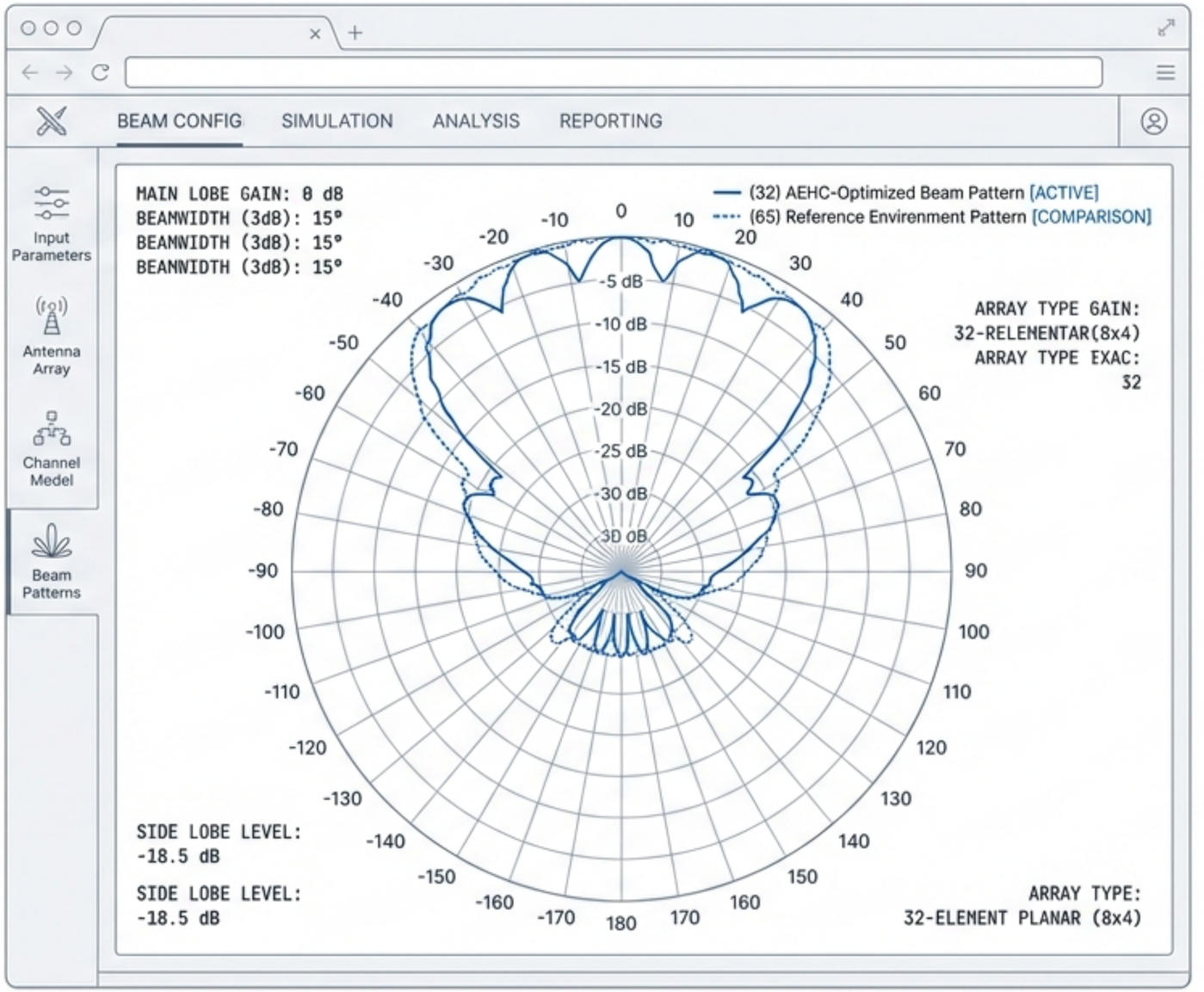
v1.0 // SPECIFICATION

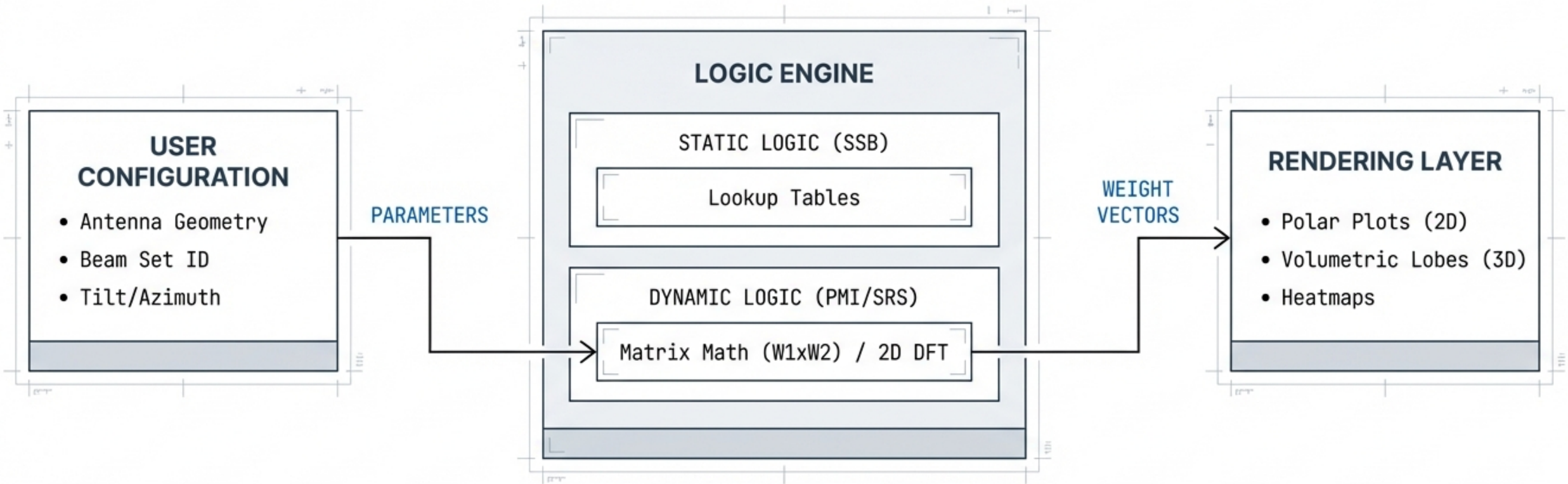
5G Beamforming Simulation

Digital Twin Specification: Configuration Inputs, Mathematical Models, and Pattern Definitions.



NotebookLM

SYSTEM ARCHITECTURE FLOW

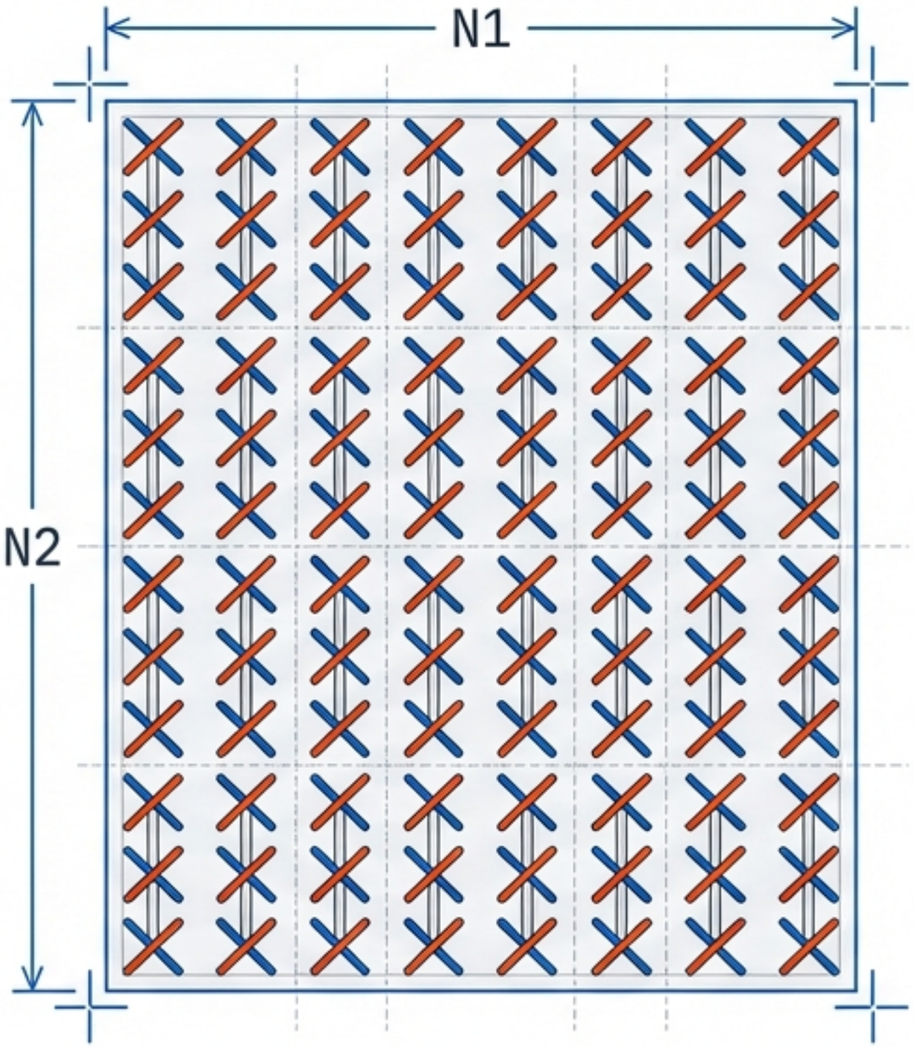


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DEFINING ANTENNA GEOMETRY

// HARDWARE INPUTS

TYPE: Active (mMIMO) vs Passive (RRH)
N1: Horizontal Beams
N2: Vertical Beams
O1/O2: Oversampling Factors
POLARIZATION: ±45°

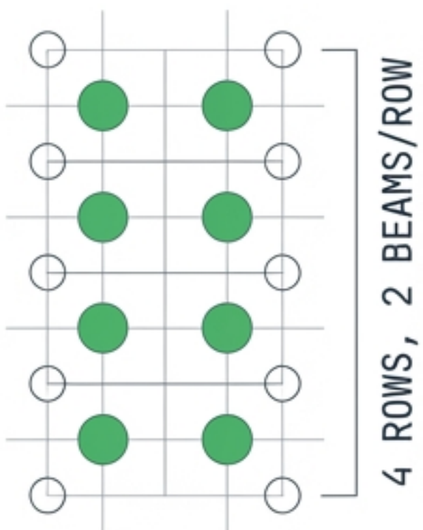


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PARSING BEAM SET NOMENCLATURE

// FORMAT: #k#l#m... (Rows Top to Bottom)

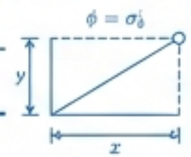


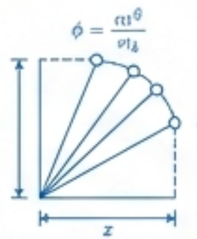


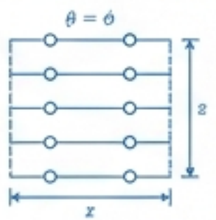
NotebookLM

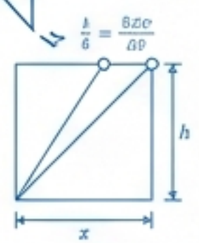
STANDARD BEAM SET PRESETS

FREQUENCY RANGE 1 (Digital)	FREQUENCY RANGE 2 (Analog)
Digital Beamforming	Analogue Beamforming
beamSet_1 (Default)	beamSetAbf_1A
beamSet_8 (1 Row, 8 Beams)	beamSetAbf_32A (mmWave)
beamSet_4_4 (2 Rows, 4 Beams)	beamSetFR2_6_6_4_2P
beamSet_2_2_2_2 (4 Rows, 2 Beams)	
beamSet_3_3_2_square	



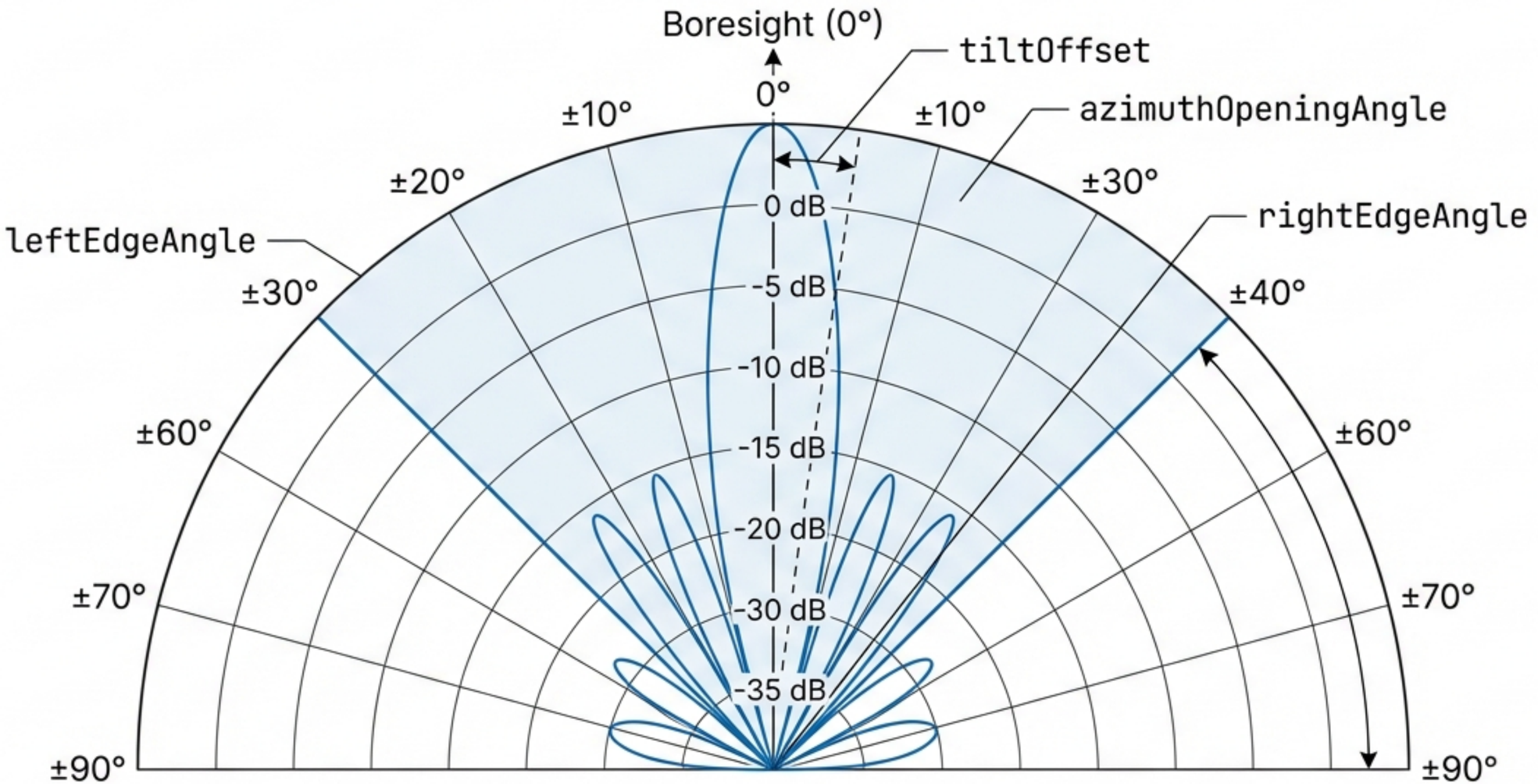






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SPATIAL CONSTRAINTS & SHAPING

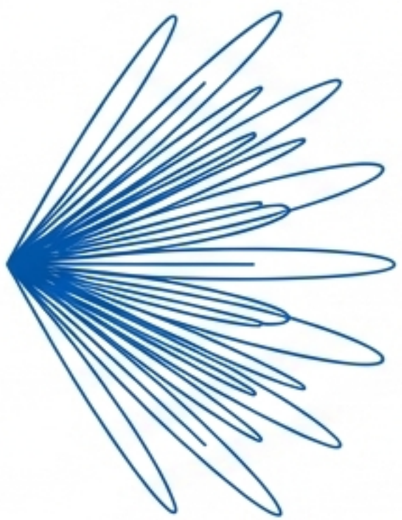


NotebookLM

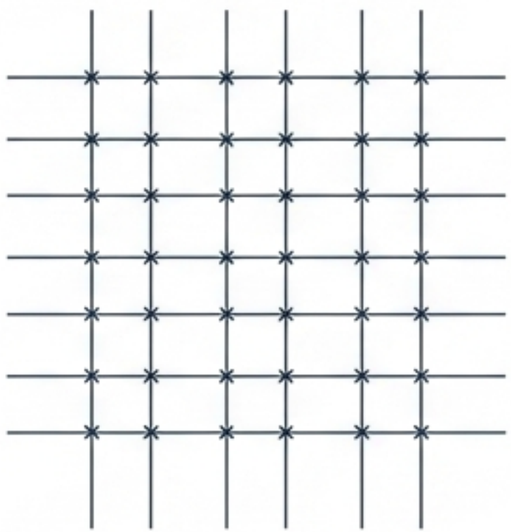
MATH MODEL: CODEBOOK BASED BEAMFORMING

PMI calculation formula

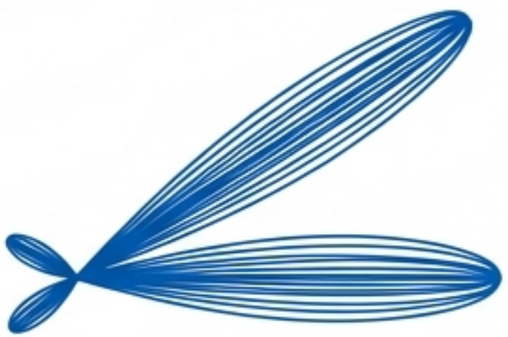
$$W = W1 \times W2$$



Beamforming Weight

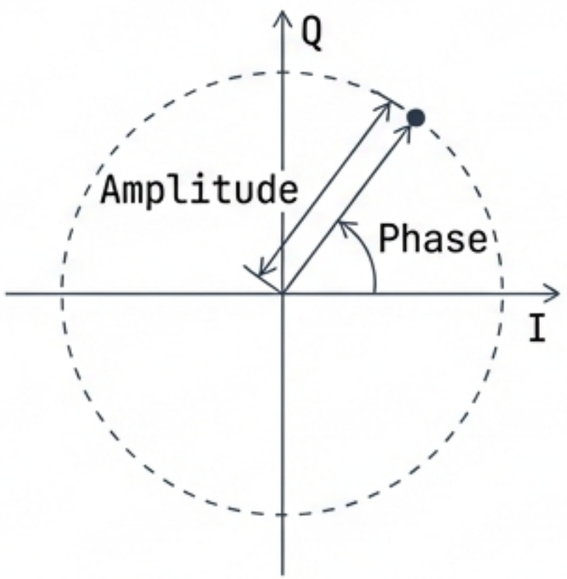


L1 Precoding Matrix



Final Focused Beam

Received Symbol



Received Symbol

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MATH MODEL: 2D DFT BEAM GENERATION

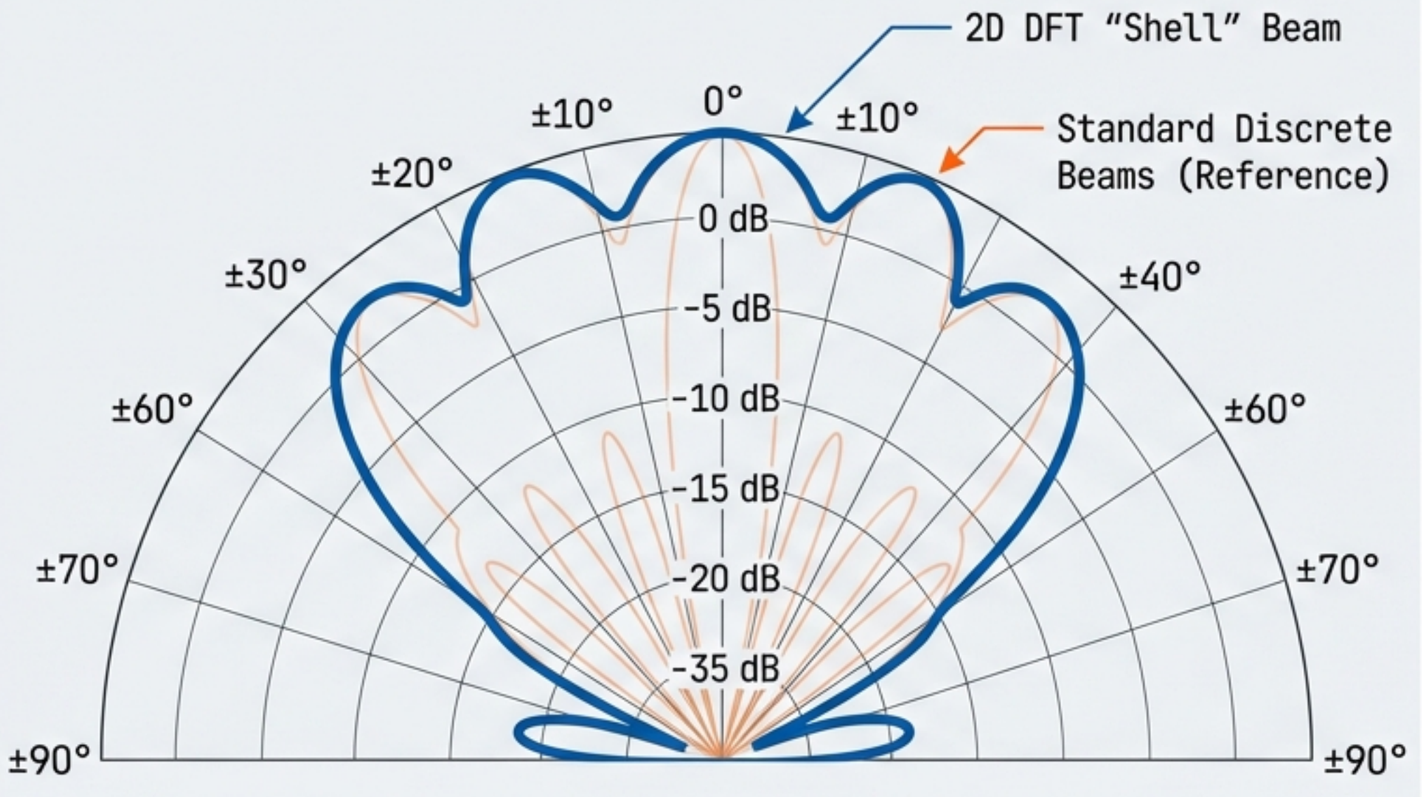
SRS (Sounding Reference Signal) Logic

THE ALGORITHM

$$Maximization = H \cdot RSRS \cdot w_k$$

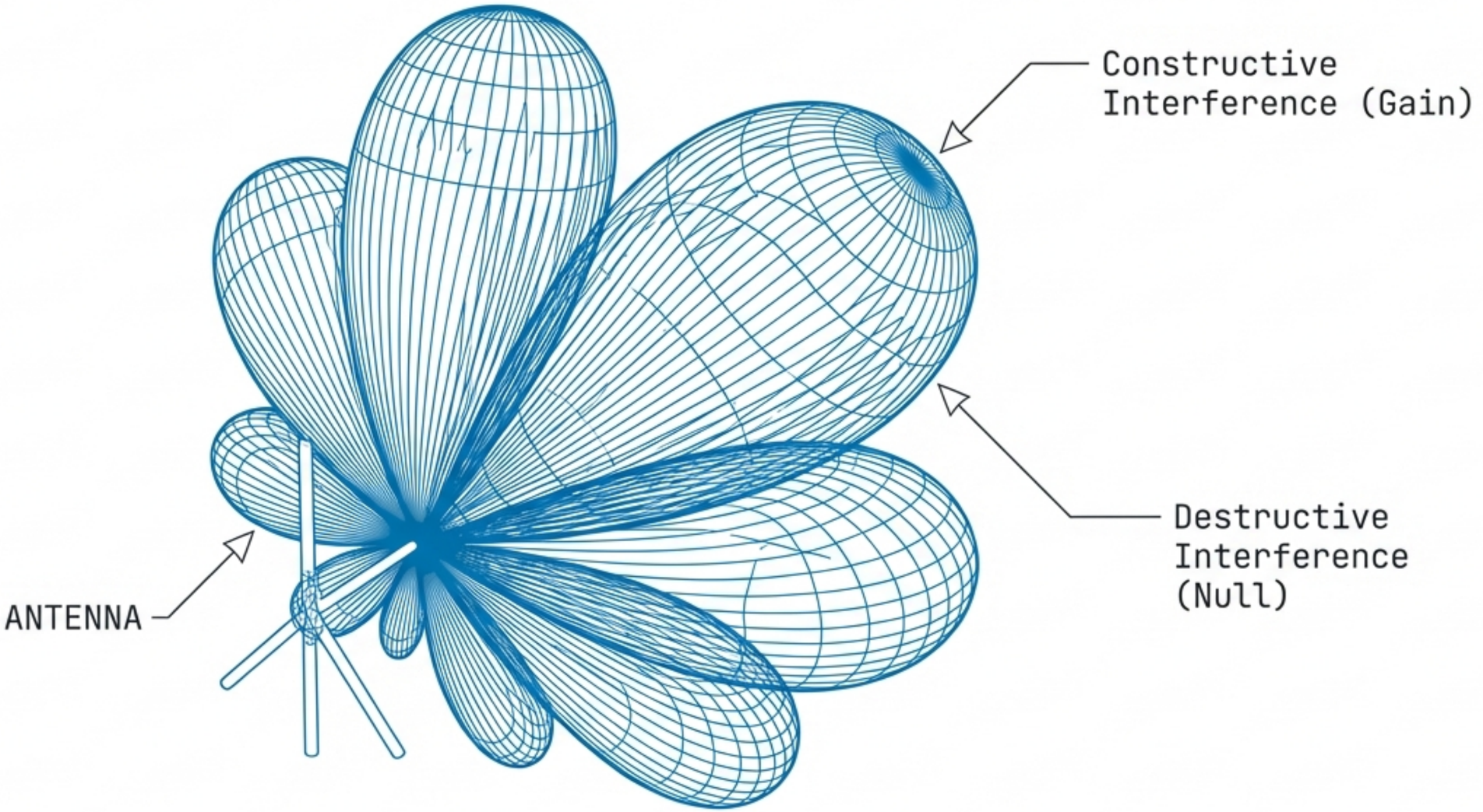
Iterate through beamforming vectors (w_k) to find highest covariance with uplink signal (SRS).

BEAM SHAPE COMPARISON



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PHYSICS ENGINE: INTERFERENCE & SHAPING



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DYNAMIC BEAM SELECTION LOGIC

START SELECTION

Initial Access (SSB)

GRID OF BEAMS

UE REPORTS TOP 2 RSRP



Tracking (CSI-RS)

MEASURE CHANNEL COEFF

CALCULATE PMI INDEX (1-64)

$$[\mathbf{H}] \cdot [\mathbf{W}_k]$$

Reciprocity (SRS)

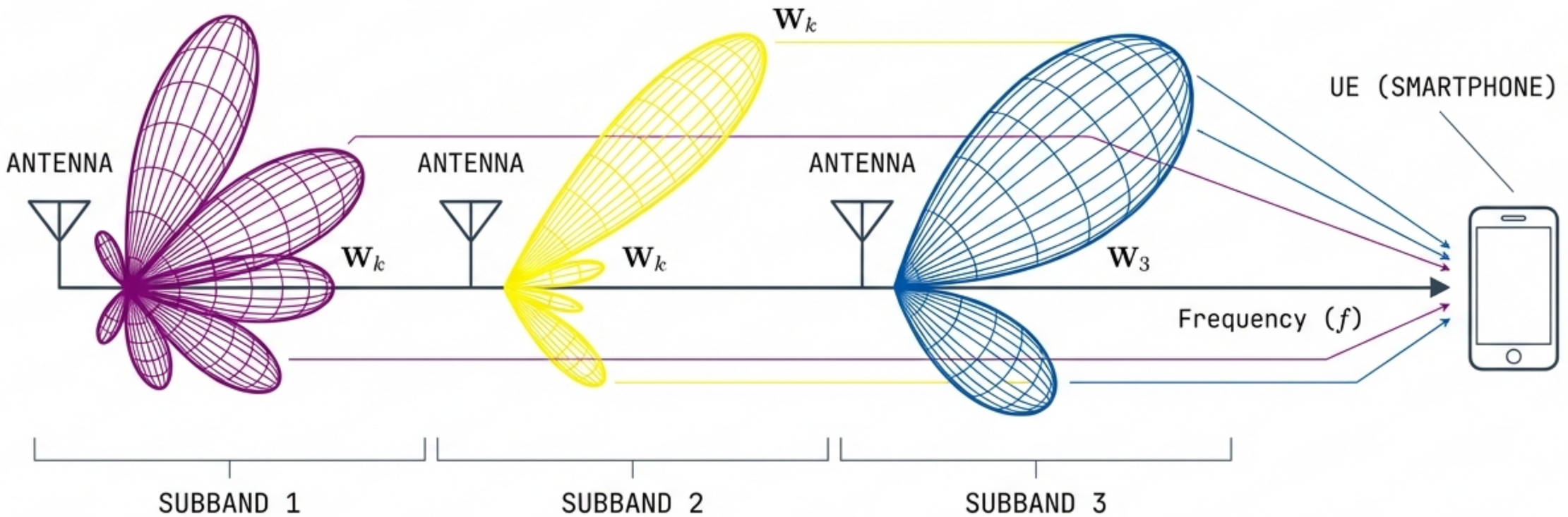
RECEIVE UPLINK SRS

CALCULATE 2D DFT WEIGHTS

$$\sum e^{-j \cdot \omega \cdot n}$$

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ADVANCED MODE: REAL-TIME SUB-BAND BEAMFORMING



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DEPLOYMENT SCENARIO RECOMMENDATIONS

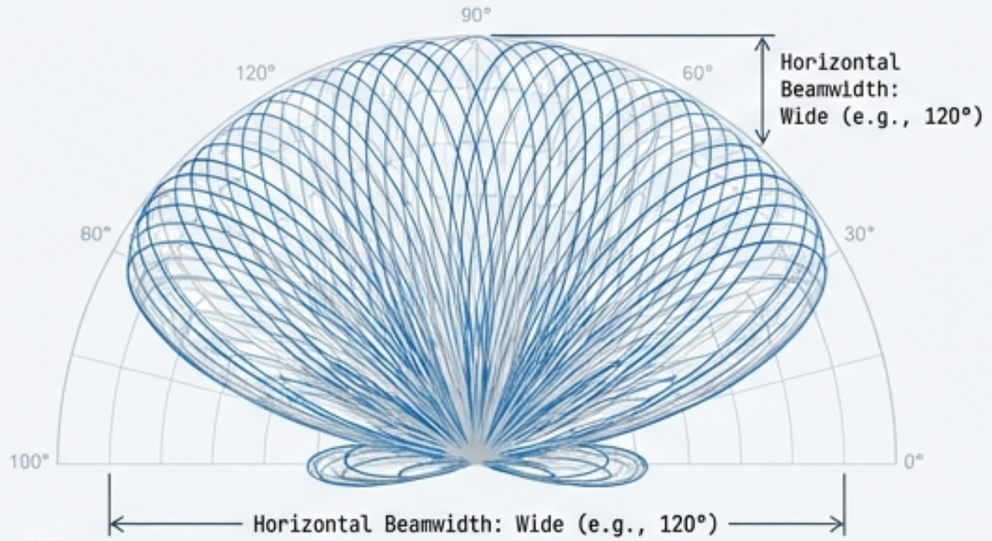
RURAL / FLAT SCENARIO

PROBLEM: WIDE AREA COVERAGE



BEAM SOLUTION: HORIZONTAL SPREAD

Focus on Horizontal Spread.



SPEC: FLAT TERRAIN OPTIMIZATION

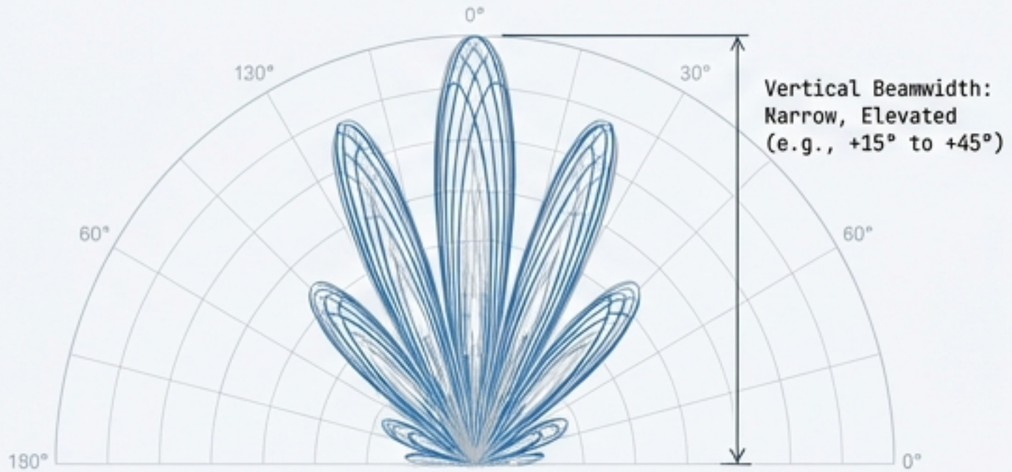
DENSE URBAN / HIGH-RISE SCENARIO

PROBLEM: VERTICAL OBSTRUCTIONS & ELEVATION



BEAM SOLUTION: VERTICAL ELEVATION

Focus on Vertical Elevation.



COMPOSITE BEAM PATTERN

OPTIMIZED FOR MAXIMUM THROUGHPUT

SPEC: HIGH-DERSITY MULTI-PATH OPTIMIZATION

MASTER PARAMETER LIST (JSON SCHEMA)

[JSON SCHEMA]

```
{
    "NRCELL": {
        "beamSet": {
            "basicBeamSet": "beamSet_2_2_2_2",
            "tiltOffset": 5,
            "azimuthOpeningAngle": 120,
            "upperEdgeAngle": 45,
            "lowerEdgeAngle": 135
        },
        "dlMimoMode": "Closed Loop Spatial Mux"
    },
    "ANTENNA": {
        "mMIMOAntArrayMode": "Top/Bottom Split",
        "columns": 8,
        "rows": 4,
        "polarization": "PlusMinus45"
    },
    "REPORTING": {
        "reportQuantity": "ssb-Index-RSRP"
    }
}
```

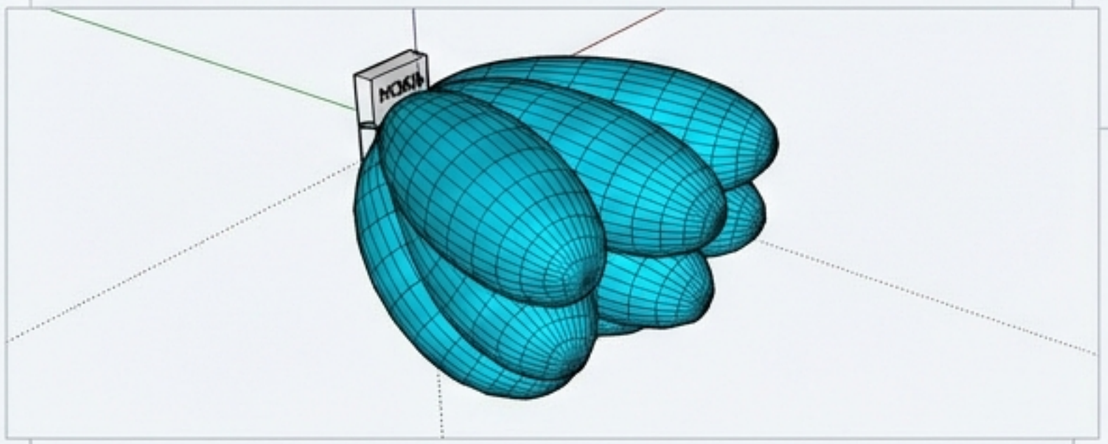
LINE NO.

SPEC: JSON PARAMETER CONFIGURATION

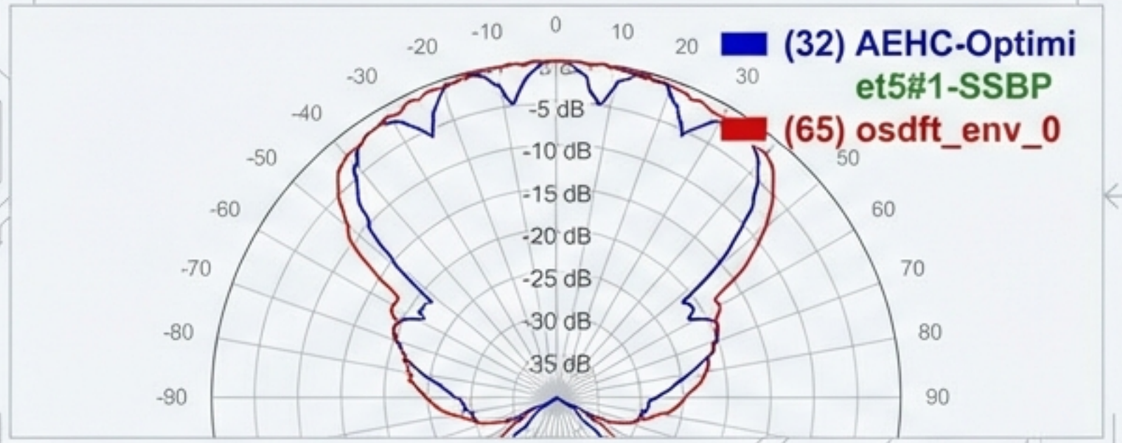
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VISUALIZATION OUTPUT SPECIFICATIONS

3D VIEWPORT



2D POLAR CUT



SPEC: VISUALIZATION DASHBOARD UI

METRICS CARD

Peak Gain: **18 dBi**

Interference Delta: **-12 dB**

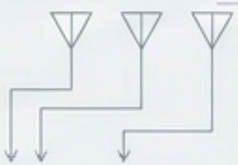
Est. Coverage Area: **2.4 sq km**

SPEC: JETBRAINS NONO

NotebookLM

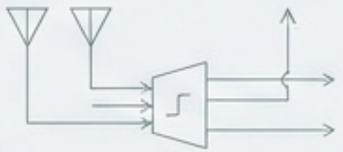
IMPLEMENTATION ROADMAP

01



GEOMETRY ENGINE

Antenna Inputs & Parsers.



02

$$\Sigma = \sum_{i=1}^{n} o_i = \begin{bmatrix} a_1 & b_1 \\ 0 & 0_1 \\ b_1 & b_2 \end{bmatrix}$$

$$\prod = \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix}$$

MATH CORE

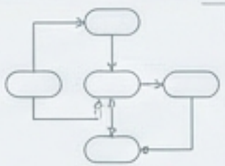
Matrix Multiplication & DFT.

$$\Sigma = \sum_{i=1}^{n} n_i (1 - \Pi).$$

$$\Pi = \begin{bmatrix} 20 & 2 & 0 \\ 90 & 1 & 0 \\ 0 & 0 & -9 \end{bmatrix}$$

$$\Pi = \prod_{i=1}^{n} \beta_i \cdot$$

03

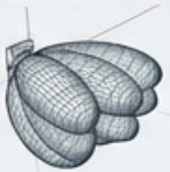


LOGIC LAYER

SSB/CSI-RS State Machine.



04



VISUALIZATION

Three.js / WebGL Rendering.

```
WebGL {
    geometry: {
        scene.addMesh(mesh);
        gpu.setAttribute("vertex");
        draw.render();
    };
    renderLoop {
        code.secretGameState();
    }
```

v1.0 RELEASE

SPEC: IMPLEMENTATION ROADMAP

NotebookLM